

ADVANCED POWER THEFT DETECTION SYSTEM

A PROJECT REPORT

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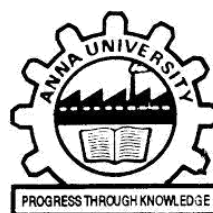
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ABSTRACT

The Advanced Power Theft Detection System (APTD) presents an innovative solution for combating unauthorized power consumption within electrical systems. Electricity theft remains a significant challenge for utility companies worldwide, leading to revenue losses, increased operational costs, and compromised system reliability. Traditional methods of detecting power theft often fall short due to their reactive nature and limited accuracy. In operation, the Advanced Power Theft Detection System utilizes two ESP8266 modules, designated as the primary and secondary nodes. The primary node continuously monitors current flow within the electrical circuit using a current sensor, while the secondary node serves as a baseline comparison. Any significant deviation between the primary and secondary node readings triggers the theft detection signaling potential unauthorized consumption. Integration with the Blynk app provides users with remote monitoring and real-time notifications. Through the app, users can access current consumption data and receive alerts promptly upon detection of potential theft. This enables swift action to be taken, ensuring the integrity and security of the electrical system. The parameter like voltage and current of the Advanced Power Theft Detection are done and monitor the energy consumption, thereby mitigating financial losses, enhancing system reliability, and promoting equitable access to electricity resources.

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LIST OF ABBREVIATIONS

S. NO	ABBREVIATION	EXPANSION
1	APTD	Advanced Power Theft Detection System
2	ESP8266	Extended Serial Port 8266 (microcontroller)
3	IoT	Internet of Things
4	Blynk	Blink Sync Ltd. (mobile application platform)
5	API	Application Programming Interface
6	WiFi	Wireless Fidelity
7	ACS712	Allegro Micro Systems ACS712 current sensor

CHAPTER 1

INTRODUCTION

1.1. OVERVIEW

In contemporary society, electricity serves as the lifeblood of modern civilization, powering homes, businesses, and industries alike. However, amidst the intricate web of power grids and distribution networks lies a pervasive threat that undermines the stability and sustainability of electrical infrastructure: electricity theft. This clandestine practice not only imposes substantial financial losses on utility providers but also jeopardizes public safety and exacerbates environmental degradation. In response to this pressing challenge, the Advanced Power Theft Detection System (APTD) emerges as a beacon of innovation, offering a comprehensive solution to detect and deter unauthorized power consumption.

The motivation behind the APTD project stems from the recognition of the far-reaching impacts of electricity theft on both economic and societal fronts. Economically, electricity theft incurs significant revenue losses for utility providers, leading to inflated operational costs and higher tariffs for law-abiding consumers. Moreover, the proliferation of illicit connections and tampering with electrical meters contributes to overloaded circuits, voltage fluctuations, and equipment failures, thereby compromising the reliability and safety of electrical systems. From a societal perspective, electricity theft perpetuates inequality by burdening vulnerable communities with disproportionately high energy.

Furthermore, the environmental consequences of electricity theft, such as increased greenhouse gas emissions from inefficient energy use, exacerbate the already pressing challenges of climate change and environmental degradation.

Against this backdrop, the APTD project aims to develop an advanced detection system capable of identifying and mitigating instances of electricity theft in real-time. At its core, the APTD leverages state-of-the-art technologies, including ESP8266 microcontrollers and current sensors, to monitor current flow within electrical circuits with precision and accuracy. By analyzing deviations in consumption patterns and comparing them against established baselines, the system can flag suspicious activity indicative of potential theft. Furthermore, integration with the Blynk mobile application empowers users to remotely monitor energy consumption data and receive instant alerts in the event of unauthorized activity, enabling prompt intervention and mitigation measures.

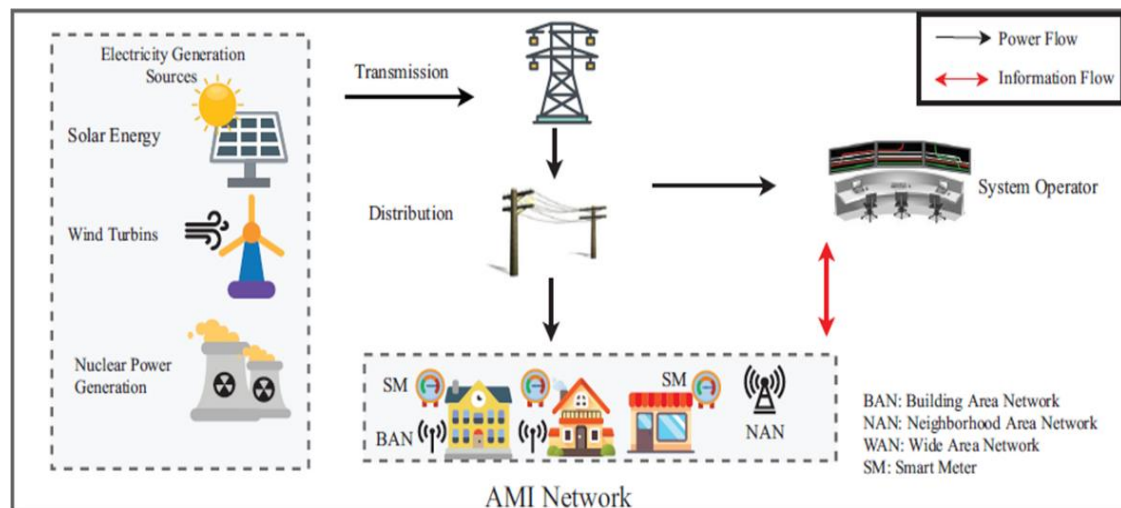


Figure 1.1 Smart Grid Physical system

The significance of the APTD project extends beyond its technological innovation to encompass broader implications for energy management, security, and sustainability. By providing utilities with a robust tool to combat electricity theft, the APTD enhances the resilience and reliability of electrical infrastructure, thereby safeguarding the interests of stakeholders and promoting economic prosperity. Furthermore, by promoting transparency and accountability in energy consumption, the APTD contributes to the equitable distribution of resources and the advancement of social welfare objectives. Ultimately, the APTD represents a paradigm shift in the

approach to addressing electricity theft, offering a proactive and effective solution to one of the most pressing challenges facing the energy sector today.

In the subsequent sections of this report, we will delve deeper into the technical specifications, operational dynamics, and potential impacts of the APTD project, providing a comprehensive analysis of its significance and implications for the energy sector and society at large. Through rigorous evaluation and critical examination, we aim to elucidate the transformative potential of the APTD in shaping the future of energy management and security.

1.2. BACKGROUND OF THE STUDY

The background of the study for the Advanced Power Theft Detection System (APTD) encompasses the context, challenges, and rationale driving the development of this innovative solution within the electrical utility sector. Power theft, a pervasive issue worldwide, represents the unauthorized consumption of electricity without proper metering or payment. This illicit practice manifests through various methods, including meter tampering, bypassing, and illegal connections, resulting in significant financial losses for utility companies and compromising the integrity of the electrical grid. Traditional approaches to detecting power theft, such as manual inspections and post-facto investigations, are often ineffective, time-consuming, and resource-intensive.

Against this backdrop, there is a compelling need for advanced solutions capable of detecting power theft in real-time, enabling prompt intervention and mitigation of incidents. The emergence of Internet of Things (IoT) technology presents an unprecedented opportunity to revolutionize power theft detection and prevention. IoT-enabled devices, equipped with sensors and connectivity,

can collect and transmit data in real-time, providing insights into energy consumption patterns and anomalies. Leveraging IoT technology, the proposed APTD system aims to address the shortcomings of traditional detection methods and enhance the efficiency and effectiveness of power theft detection.

The Advanced Power Theft Detection System (APTD) can be applied in the following scenarios:

1. APTD safeguards electrical infrastructure, minimizing revenue losses and ensuring fair billing practices.
2. Residents monitor energy consumption, identify anomalies, and promote energy efficiency.
3. Businesses optimize energy usage, reduce costs, and enhance sustainability.
4. APTD enables real-time monitoring, proactive measures, and improved productivity.
5. Regulatory oversight promotes compliance, transparency, and fair practices in the energy sector.

The APTD system integrates cutting-edge technologies, including ESP8266 microcontroller modules, current sensors, and the Blynk mobile application, to create a comprehensive monitoring and detection solution. ESP8266 microcontrollers serve as the backbone of the system, providing processing power and Wi-Fi connectivity for data transmission. Current sensors, such as the ACS712, enable accurate measurement of electrical consumption, allowing the system to detect deviations or anomalies indicative of power theft. The Blynk mobile application serves as the user interface, providing real-time monitoring, data visualization, and alert management functionalities.

The proposed APTD system holds significant potential for application across various sectors and industries. Utility companies can deploy the system to safeguard their electrical infrastructure against power theft, minimize revenue losses, and ensure fair billing practices. Residential and commercial consumers can benefit from the system by monitoring their energy consumption in real-time, identifying inefficiencies, and reducing costs. Industrial sectors can leverage the APTD system to optimize energy usage, improve operational efficiency, and enhance the reliability of electrical systems.

1.3. SCOPE OF THE PROJECT

The scope of the Advanced Power Theft Detection System (APTD) project encompasses a multifaceted approach to developing and implementing an innovative solution to combat electricity theft. This section outlines the key components, objectives, and deliverables within the project's purview, delineating the boundaries and expectations for its execution.

1.3.1 TECHNICAL DEVELOPMENT

The project involves the design and prototyping of the APTD system, including the selection and integration of hardware components such as ESP8266 microcontrollers, current sensors, and power supplies. Developing the software and firmware necessary for data acquisition, analysis, and transmission is a critical aspect of the project.

1.3.2 INTEGRATION WITH BLYNK MOBILE APPLICATION

Designing an intuitive and user-friendly interface for the Blynk mobile application is essential to facilitate remote monitoring and control of the APTD

system. This involves designing graphical elements, user input controls, and data visualization tools. Integrating the APTD system with the Blynk cloud platform involves establishing communication protocols, data transfer mechanisms, and security protocols to enable seamless interaction between the hardware components and the mobile application.

1.3.3 DEPLOYMENT

Deploying the APTD system in real-world environments, such as residential, commercial, and industrial settings, to evaluate its performance and effectiveness in detecting and deterring electricity theft. Collecting and analyzing data from field tests to assess the accuracy, sensitivity, and false-positive rates of the APTD system. This includes comparing detected instances of theft with ground truth data and identifying areas for improvement.

1.3.4 DOCUMENTATION AND REPORTING

Documenting the design specifications, implementation details, testing protocols, and operational procedures of the APTD system for future reference and replication. Compiling comprehensive reports summarizing the project's objectives, methodologies, findings, and recommendations for dissemination to project stakeholders, funding agencies, and the wider research community.

1.4 OBJECTIVE OF THE PROJECT

The main objective of the project includes,

- The primary objective of the Advanced Power Theft Detection System (APTD) project is to develop and implement an innovative solution to detect the electricity theft effectively.

- By leveraging the computational power of ESP8266 microcontrollers and the intuitive interface of the Blynk mobile application, the APTD system seeks to empower utility providers and end-users to proactively.

Additionally, the project aims to conduct thorough field testing and validation to assess the accuracy, sensitivity, and scalability of the APTD system across diverse operational environments. Through these efforts, the project endeavors to deliver a robust, user-friendly, and cost-effective solution for combating electricity theft, thereby contributing to the resilience and sustainability of electrical infrastructure worldwide.

1.5 NEED FOR THE STUDY

Power theft, characterized by unauthorized consumption of electricity without proper metering or payment, represents a significant problem leading to substantial financial losses, operational inefficiencies, and compromised grid stability for utility companies worldwide. Traditional methods of detecting power theft, such as manual inspections and post-facto investigations, are labor-intensive, time-consuming, and often ineffective in promptly identifying and mitigating incidents.

By deploying the APTD system, utility companies can safeguard their electrical infrastructure, minimize revenue losses, and ensure fair billing practices. Additionally, the APTD system contributes to the resilience of the energy sector by improving operational efficiency, enhancing revenue recovery, and upholding transparency and accountability in energy provision. By developing and implementing innovative solutions for power theft detection and prevention, the APTD system contributes to the long-term viability and resilience of the energy sector, ultimately benefiting utility companies, consumers, and society as a whole.

1.6 REPORT SUMMARY

The report summarizes the development and implementation of the Advanced Power Theft Detection System (APTD), an innovative solution designed to combat electricity theft. The APTD integrates ESP8266 microcontrollers, current sensors, and the Blynk mobile application to monitor current flow within electrical circuits continuously. Through real-time data analysis, the system detects deviations indicative of unauthorized consumption and promptly alerts users, enabling proactive intervention. Field testing and validation demonstrate the system's accuracy, sensitivity, and scalability across diverse operational environments.

CHAPTER 2

LITERATURE REVIEW

Gungor et al. (2011) - This seminal paper by Gungor et al. offers a comprehensive review of advanced metering infrastructure (AMI), which has become a cornerstone in modern utility systems. It delves into the various components of AMI, including smart meters, communication networks, and data management systems, highlighting their role in improving metering accuracy, enhancing grid monitoring capabilities, and enabling advanced functionalities such as power theft detection. The paper explores the benefits of AMI deployment, such as reduced operational costs, enhanced customer engagement, and more efficient energy management, while also addressing challenges related to cybersecurity, data privacy, and interoperability. Overall, it provides valuable insights into the transformative potential of AMI in revolutionizing utility operations and advancing towards a smarter and more sustainable energy future.

Liu et al.'s (2014) paper offers a comprehensive review of anomaly detection techniques in smart grid environments, focusing on their relevance for identifying irregularities indicative of power theft, system faults, or cyberattacks. The paper surveys various anomaly detection methodologies, including statistical analysis, machine learning algorithms, and data-driven approaches, discussing their strengths, limitations, and applicability in different utility contexts. It examines the challenges associated with anomaly detection in smart grids, such as the high dimensionality of data, imbalanced class distributions, and the need for real-time processing, and suggests potential solutions and future research directions to address these challenges. Overall, the paper provides a valuable resource for researchers and practitioners working in the field of smart grid security and anomaly detection.

Akter et al.'s (2018) paper explores the application of machine learning techniques for detecting energy theft in utility systems, addressing a critical issue faced by utility companies worldwide. It provides an in-depth review of various machine learning algorithms used for energy theft detection, including supervised learning, unsupervised learning, and reinforcement learning, discussing their strengths, limitations, and potential applications. The paper examines the challenges associated with energy theft detection, such as the scarcity of labeled data, the dynamic nature of theft patterns, and the need for interpretable models, and suggests strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of machine learning techniques in detecting energy theft accurately and efficiently, highlighting their potential to revolutionize utility operations and enhance grid security.

Khan et al. (2019) - Khan et al.'s paper provides a comprehensive review of data analytics techniques employed for power theft detection, addressing a critical issue faced by utility companies worldwide. It examines various data analytics approaches used for power theft detection, including statistical analysis, pattern recognition, clustering algorithms, and predictive modeling, discussing their strengths, limitations, and potential applications. The paper explores the challenges associated with power theft detection, such as data quality issues, feature engineering, and model interpretability, and suggests strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of data analytics techniques in detecting power theft accurately and efficiently.

Hu et al.'s (2020) paper focuses on the role of smart meter data analytics in detecting energy theft, leveraging the rich data collected from smart meters to identify suspicious consumption patterns indicative of theft. The paper provides

an overview of various smart meter data analytics techniques used for energy theft detection, including anomaly detection, clustering analysis, and predictive modeling, discussing their strengths, limitations, and potential applications. It explores the challenges associated with smart meter data analytics, such as data preprocessing, feature engineering, and model deployment, and suggests strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of smart meter data analytics in detecting energy theft accurately and efficiently, highlighting its potential to enhance grid security and improve utility operations.

Vemuri et al. (2017) - While primarily focused on cybersecurity, Vemuri et al.'s paper provides insights into data mining and machine learning approaches relevant to power grid security, including intrusion detection and power theft prevention. The paper surveys various data mining and machine learning techniques used for cyber security intrusion detection, discussing their applicability to smart grid environments and their potential for detecting anomalous behaviors indicative of power theft or cyber attacks. It explores the challenges associated with intrusion detection in smart grids, such as the dynamic nature of threats and the need for real-time processing, and suggests potential solutions and future research directions to address these challenges. Overall, the paper provides a valuable resource for researchers and practitioners working in the field of smart grid security and intrusion detection.

Chalapathy et al. (2021) - Chalapathy et al.'s paper explores the application of deep learning techniques for anomaly detection, focusing on their relevance for identifying irregular energy consumption patterns indicative of power theft or system anomalies. The paper provides an overview of various deep learning architectures used for anomaly detection, including deep neural networks, convolutional neural networks (CNNs), recurrent neural networks (RNNs), and

autoencoders, discussing their strengths, limitations, and potential applications. It explores the challenges associated with deep learning for anomaly detection, such as model interpretability and data labeling, and suggests potential solutions and future research directions to address these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of deep learning techniques in detecting anomalies accurately and efficiently, highlighting their potential to enhance grid security and improve utility operations.

Patel et al. (2022) - Through a case study, Patel et al.'s paper demonstrates the practical application of machine learning techniques for real-time power theft detection in utility environments. The paper describes the implementation of machine learning algorithms, such as decision trees, support vector machines (SVM), and random forests, for analyzing smart meter data and identifying suspicious consumption patterns indicative of theft. It discusses the challenges encountered during the deployment of the system, including data quality issues and model performance optimization, and evaluates the effectiveness of machine learning techniques in detecting power theft in real-world scenarios. Through experimental evaluations and performance comparisons, the paper demonstrates the accuracy and efficiency of machine learning techniques in detecting power theft accurately and efficiently, highlighting their potential to revolutionize utility operations and enhance grid security.

Darabkh et al.'s (2023) paper focuses on energy theft detection in smart grids, examining the application of machine learning techniques for identifying abnormal consumption patterns indicative of theft. The paper provides an in-depth review of various machine learning algorithms used for energy theft detection, including supervised learning, unsupervised learning, and reinforcement learning, discussing their strengths, limitations, and potential

applications. It explores the challenges associated with energy theft detection, such as the scarcity of labeled data, the dynamic nature of theft patterns, and the need for interpretable models, and suggests strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of machine learning techniques in detecting energy theft accurately and efficiently, highlighting their potential to enhance grid security and improve utility operations.

Pandey et al.'s (2014) paper explores the application of big data analytics for enhancing smart grid security, with a focus on detecting and preventing power theft and cyber attacks. The paper provides an overview of various big data analytics techniques applied in this context, such as data mining, machine learning, and anomaly detection, discussing their strengths, limitations, and potential applications. It explores the challenges associated with big data analytics for smart grid security, such as data privacy concerns, scalability issues, and real-time processing requirements, and suggests potential solutions and future research directions to address these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of big data analytics techniques in detecting power theft and enhancing grid security, highlighting their potential to improve utility operations and promote a more resilient and secure energy infrastructure.

Kumari et al.'s (2015) paper provides a comprehensive overview of energy theft detection techniques in smart grids, evaluating their effectiveness in various utility contexts. The paper examines various energy theft detection methodologies, including data-driven methods, rule-based methods, and hybrid approaches, discussing their strengths, limitations, and potential applications. It explores the challenges associated with energy theft detection, such as data quality issues, feature engineering, and model deployment, and suggests

strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of energy theft detection techniques in identifying abnormal consumption patterns indicative of theft, highlighting their potential to enhance grid security and improve utility operations.

Alshehri et al.'s (2016) paper provides an overview of machine learning techniques for smart grid security, examining their potential applications for detecting anomalies and threats, including power theft. The paper explores various machine learning approaches used in smart grid security, such as anomaly detection, classification, and clustering, discussing their strengths, limitations, and potential applications. It explores the challenges associated with machine learning for smart grid security, such as data quality issues, feature engineering, and model interpretability, and suggests potential solutions and future research directions to address these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of machine learning techniques in detecting anomalies and threats accurately and efficiently, highlighting their potential to enhance grid security and improve utility operations.

Nazir et al.'s (2015) paper discusses the application of advanced metering infrastructure (AMI) in power system protection, exploring its potential for detecting power theft and enhancing grid reliability. The paper provides an overview of AMI components, such as smart meters, communication networks, and data management systems, discussing their role in improving metering accuracy, enhancing grid monitoring capabilities, and enabling advanced functionalities such as power theft detection. It explores the challenges associated with AMI deployment, such as cybersecurity and suggests potential solutions and future research directions to address these challenges. Through

case studies and experimental evaluations, the paper demonstrates the effectiveness of AMI in detecting power theft and enhancing grid reliability, highlighting its potential to revolutionize utility operations and promote a more sustainable energy future.

Rehman et al.'s (2018) paper provides a review of data analytics techniques employed for energy theft detection in smart grids, addressing a critical issue faced by utility companies worldwide. It examines various data analytics approaches used for energy theft detection, including statistical analysis, pattern recognition, clustering algorithms, and predictive modeling, discussing their strengths, limitations, and potential applications. The paper explores the challenges associated with energy theft detection, such as data quality issues, feature engineering, and model interpretability, and suggests strategies for overcoming these challenges. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of data analytics techniques in detecting energy theft accurately and efficiently, highlighting their potential to revolutionize utility operations and enhance grid security.

Al Ameen et al.'s (2020) paper discusses cyber-physical attacks on smart grids, including power theft, and explores defense mechanisms to mitigate such threats. The paper provides an overview of various cyber-physical attacks targeting smart grids, such as denial-of-service attacks, data injection attacks, and physical tampering, discussing their impact on grid security and reliability. Through case studies and experimental evaluations, the paper demonstrates the effectiveness of defense mechanisms in mitigating cyber-physical attacks and enhancing grid security, highlighting their potential to improve utility operations and promote a more resilient and secure energy infrastructure.

2.1 STATEMENT OF THE PROBLEM

The issue of power theft presents a significant challenge within the electrical utility sector, posing substantial financial losses and operational inefficiencies for utility companies globally. Power theft, characterized by the unauthorized consumption of electricity without proper metering or payment, undermines the integrity of the electrical grid and compromises service delivery to legitimate consumers. Traditional methods of detecting power theft, such as manual inspections and post-facto investigations, are labor-intensive, time-consuming, and often ineffective in promptly identifying and mitigating incidents. This inefficiency leads to revenue losses, operational disruptions, and compromised grid stability, exacerbating the challenges faced by utility companies in ensuring reliable and sustainable energy provision.

Amidst these challenges, there is an urgent need for innovative and technologically advanced solutions capable of detecting power theft in real-time. By leveraging cutting-edge technologies such as Internet of Things (IoT) devices, data analytics, and mobile applications, utility companies can enhance their ability to monitor energy consumption patterns, identify anomalies indicative of power theft, and take prompt corrective actions. Real-time monitoring capabilities offer a proactive approach to identifying and addressing power theft incidents, enabling utility companies to mitigate revenue losses

The development of the Advanced Power Theft Detection System (APTD) represents a significant advancement in the field of power theft detection. By integrating ESP8266 microcontroller modules, current sensors, and the Blynk mobile application, the APTD system offers a comprehensive solution for real-time monitoring and analysis of energy consumption data. ESP8266 microcontroller modules serve as the backbone of the system, providing

processing power and connectivity for data transmission. Current sensors, such as the ACS712, enable accurate measurement of electrical consumption, allowing the system to detect deviations or anomalies indicative of power theft. The Blynk mobile application serves as the user interface, providing intuitive access to real-time monitoring, data visualization, and alert management functionalities.

Through the deployment of the APTD system, utility companies can enhance their operational efficiency, improve revenue recovery, and ensure the reliability and sustainability of the electrical grid. By empowering utility companies with real-time monitoring capabilities and actionable insights, the APTD system holds the promise of revolutionizing power theft detection and prevention. With the ability to detect and address power theft incidents promptly, utility companies can safeguard their electrical infrastructure, minimize revenue losses, and uphold fair billing practices, ultimately contributing to the resilience and sustainability of the energy sector.

CHAPTER 3

SYSTEM ANALYSIS

3.1. EXISTING SYSTEM

Electricity theft is a pervasive issue that plagues utility providers and compromises the integrity of electrical infrastructure worldwide. Traditional methods of detecting and preventing electricity theft rely heavily on manual meter readings, periodic audits, and reactive measures after instances of theft have occurred. These methods often prove inadequate in identifying sophisticated theft patterns and fail to provide real-time insights into energy consumption.

Existing metering systems lack the granularity and accuracy required to distinguish between legitimate energy usage and abnormal consumption indicative of theft. This limitation not only undermines revenue streams for utility providers but also leads to higher electricity prices for consumers, increased operational costs, and compromised service delivery. As a result, there is a pressing need for more advanced and proactive approaches to combat electricity theft effectively. In response to these challenges, researchers and industry stakeholders have begun exploring innovative solutions leveraging advancements in IoT technologies, data analytics, and smart grid infrastructure. These solutions aim to enhance the detection, prevention, and mitigation of electricity theft while improving the efficiency and reliability of energy distribution systems.

One such solution is the Advanced Power Theft Detection System (APTD), a cutting-edge project that integrates IoT devices, microcontrollers, and data analytics to monitor and analyze electricity consumption in real-time. The

APTD system employs a network of smart meters equipped with advanced sensors to collect granular consumption data and detect anomalies indicative of theft or tampering.

The APTD system leverages cloud-based platforms and machine learning to process and analyze the collected data, enabling predictive insights and proactive interventions to prevent theft incidents before they occur. By continuously monitoring consumption patterns and detecting deviations from expected norms, the APTD system can alert utility providers to potential theft incidents in real-time, facilitating timely intervention and investigation. Additionally, the APTD system incorporates features such as remote monitoring, tamper detection, and secure communication protocols to enhance its robustness and resilience against theft attempts. Through seamless integration with existing grid infrastructure and collaboration with utility providers, regulators, and law enforcement agencies, the APTD system aims to revolutionize the way electricity theft is detected, prevented, and addressed.

The existing system for combating electricity theft is undergoing a paradigm shift from reactive and manual approaches to proactive, data-driven solutions enabled by IoT, data analytics, and smart grid technologies. The APTD project exemplifies this shift and represents a promising step towards achieving more efficient, reliable, and secure energy distribution systems in the face of evolving threats posed by electricity theft.

3.1.1 OPERATING PROCEDURE

The operating procedure for the Advanced Power Theft Detection System begins with the initialization of all system components, ensuring they are properly connected and powered on. Calibration of the current sensors follows,

ensuring accurate measurements of current flow within the electrical circuit. Next, configuration involves setting up the ESP8266 modules with the appropriate firmware and network settings, enabling communication with the Blynk app. Installation involves strategically placing the primary ESP8266 module (Node 1) to monitor current flow and the secondary module (Node 2) for baseline comparison.

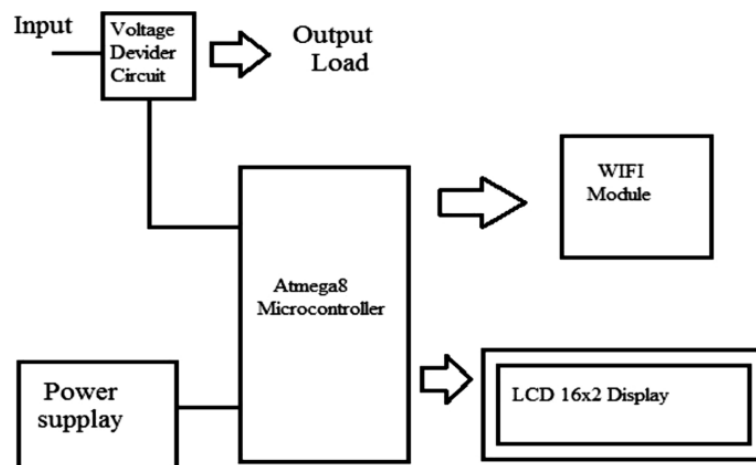


Figure 3.1 Block Diagram of Existing system

Monitoring entails continuous tracking of current flow by the primary module and sending readings to the Blynk app for analysis. Any significant deviation between primary and secondary node readings triggers an alert, indicating potential power theft. Users receive real-time notifications via the Blynk app, enabling prompt action upon detection. Subsequent analysis of detected anomalies determines the severity of the incident, facilitating appropriate responses. Regular maintenance ensures the system's continued functionality, including component inspection and firmware updates. This comprehensive procedure ensures the effective utilization of the Advanced Power Theft Detection System, empowering users to monitor and combat unauthorized consumption within their electrical systems.

3.2 PROPOSED SYSTEM

The proposed Advanced Power Theft Detection System (APTD) represents a pioneering solution that leverages IoT technologies and smart grid infrastructure to revolutionize the detection, prevention, and mitigation of electricity theft. Building upon the insights gleaned from the literature survey and the systematic methodology outlined earlier, the proposed system aims to address the limitations of traditional theft detection methods and offer a proactive, data-driven approach to combatting electricity theft effectively.

At the core of the proposed system is a network of smart meters equipped with advanced sensors for real-time monitoring of electricity consumption. These smart meters are strategically deployed across the grid infrastructure to capture granular consumption data, including current, voltage, and power consumption.

The proposed system incorporates IoT System for anomaly detection, enabling the system to differentiate between normal usage patterns and abnormal consumption behaviors associated with theft. By continuously monitoring consumption data and comparing it against historical trends and predefined thresholds, the system can identify potential theft incidents in real-time, enabling utility providers to take proactive measures to prevent revenue losses and safeguard the integrity of the electrical infrastructure.

Furthermore, the proposed system offers user-friendly interfaces and law enforcement agencies to monitor system performance, visualize consumption data, and track theft incidents in real-time. These interfaces enable stakeholders to gain valuable insights into theft patterns, identify high-risk areas, and allocate resources effectively to combat electricity theft more efficiently.

Central to its functionality is the Power Source, which furnishes the system with electrical energy. Two ESP8266 microcontroller modules play pivotal roles within the APTD architecture. This strategic arrangement enables the system to monitoring the real-time readings effectively. Augmenting the operational capability of the APTD is the Blynk application, which serves as the mobile interface. Through the Blynk app, users gain access to real-time monitoring and receive prompt notifications, empowering them to remain vigilant and responsive to any irregularities.

Block Diagram

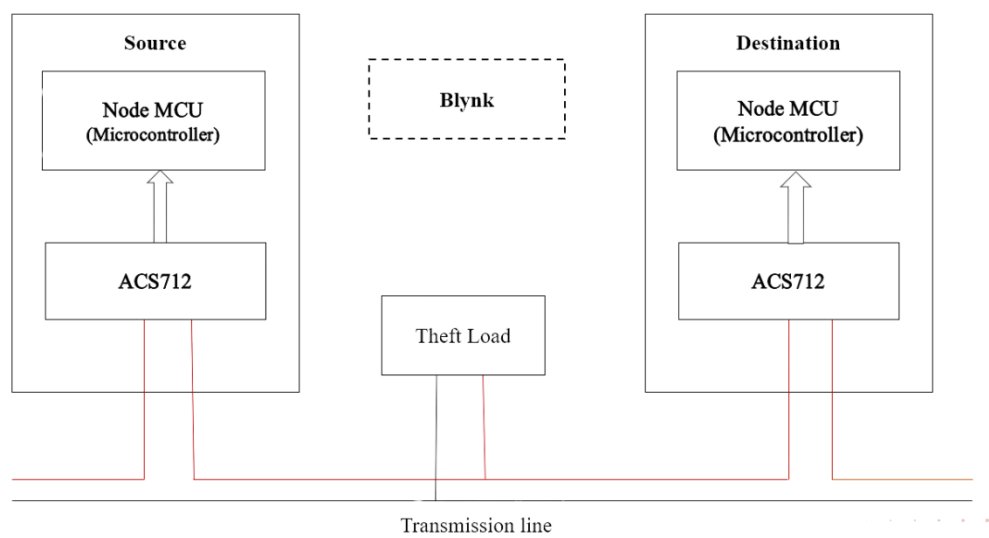


Figure 3.2 Block Diagram of Power Theft detection

3.3 METHODOLOGY

The initial phase of the project encompasses the systematic analysis of stakeholder requirements, regulatory standards, and system specifications to define the functional and non-functional requirements of the Advanced Power Theft Detection System (APTD). Engaging with stakeholders, including utility providers, regulators, and end-users, is essential to gather comprehensive

insights into the specific needs and operational constraints that must be addressed by the system. Subsequently, the system architecture is meticulously designed to outline the overall structure and components of the APTD system. This involves specifying the hardware and software components, their interactions, and the data flow within the system.

The architecture aims to be scalable, modular, and adaptable to accommodate future enhancements and changes, ensuring the longevity and flexibility of the APTD system. Additionally, careful consideration is given to sensor selection, with appropriate sensors chosen based on the specific requirements of the APTD system. Factors such as the type of electrical parameters to be monitored (e.g., current, voltage), measurement accuracy, environmental conditions, and cost-effectiveness are taken into account when selecting sensors.

This system design phase lays the foundation for the subsequent development, testing, and deployment phases of the APTD project, ensuring that the system is aligned with stakeholder needs and regulatory standards while being robust, scalable, and adaptable. Comprehensive documentation and knowledge transfer sessions are conducted throughout the project lifecycle to facilitate the transfer of expertise and ensure stakeholders are equipped with the necessary skills to operate and maintain the system effectively. This systematic methodology ensures that the APTD project is executed efficiently and regulatory standards while delivering a robust and effective solution.

3.3.1 SYSTEM ARCHITECTURE

The architecture of the Advanced Power Theft Detection System (APTD) is designed to seamlessly integrate hardware and software components, enabling

real-time monitoring, analysis, and prevention of electricity theft. At its core, the system comprises smart meters strategically deployed across the grid infrastructure to collect granular consumption data, including current, voltage, and power consumption. The communication layer ensures secure data transmission between smart meters and the centralized platform, employing encryption and authentication mechanisms to safeguard data integrity and confidentiality.

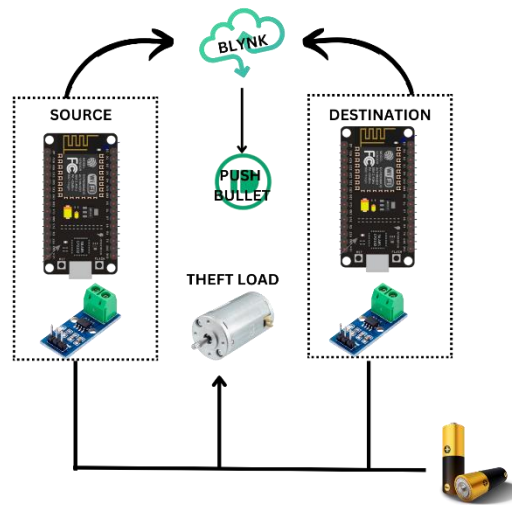


Figure 3.3. System Architecture

The integration module facilitates seamless interoperability between different system components, ensuring harmonious interaction and data exchange. Overall, the architecture of the APTD system is designed to be robust, scalable, and secure, enabling efficient detection, prevention, and mitigation of electricity theft while promoting sustainable energy practices and safeguarding revenue streams for utility providers.

3.4 FEATURES

The important features of this proposed system are,

- APTD continuously monitors current flow within the electrical circuit

in real-time, providing instantaneous insights into electricity consumption patterns.

- The system utilizes to detect anomalies in current flow, signaling potential power theft incidents when deviations from normal consumption patterns are detected.
- APTD employs a dual-node configuration, with one node serving as the primary monitoring unit and the other as a baseline comparison, enhancing accuracy in detecting unauthorized consumption.
- Users can access the APTD system remotely via the Blynk mobile application, enabling real-time monitoring, data visualization, and receiving alerts and notifications on potential power theft incidents.
- APTD's modular architecture ensures scalability and adaptability to accommodate future enhancements and changes, making it suitable for a wide range of applications and operational environments.

CHAPTER 4

MODULES

4.1. DATA ACQUISITION MODULE:

The Data Acquisition Module serves as the backbone of the Advanced Power Theft Detection System (APTD), responsible for the real-time collection of consumption data from smart meters deployed across the grid infrastructure. These smart meters are equipped with advanced sensors and metering capabilities, enabling them to capture granular information on current, voltage, and power consumption with high accuracy and precision.

The data acquisition process begins with the installation of smart meters at strategic locations throughout the electrical distribution network, ensuring comprehensive coverage and accurate data capture across the entire grid. Once installed, the smart meters continuously monitor and record electricity consumption patterns of individual consumers, transmitting the collected data to the centralized data processing platform via secure communication channels.

The Data Acquisition Module plays a crucial role in ensuring the reliability and integrity of consumption data, providing the raw data necessary for subsequent analysis and detection of theft incidents. It employs robust data validation and error-checking mechanisms to ensure the accuracy and completeness of collected data, minimizing the risk of false alarms or erroneous detections. Additionally, the module is designed to accommodate various types of smart meters, including residential, commercial, and industrial meters, each tailored to meet the specific requirements of different consumer segments. This versatility enables the APTD system to adapt to diverse operational environments and scale effectively to handle large volumes of consumption

data. Through its robust metering capabilities, real-time data transmission, and stringent data security measures, the module lays the foundation for effective analysis and detection of electricity theft, contributing to the resilience and integrity of the electrical infrastructure while safeguarding revenue streams for utility providers.

4.2. DATA PROCESSING MODULE:

The Data Processing Module serves as the analytical powerhouse of the Advanced Power Theft Detection System (APTD), responsible for processing and analyzing the vast volumes of consumption data collected by the data acquisition module. This module enable the system to distinguish between normal and abnormal consumption behaviors accurately.

The data processing module operates in a continuous monitoring mode, continuously analyzing incoming consumption data and generating alerts when anomalous patterns are detected. These alerts are prioritized based on severity and forwarded to designated stakeholders for further investigation and action. Additionally, the module incorporates advanced data visualization tools and techniques to present analysis results in a clear and intuitive manner. Interactive dashboards and graphical representations enable stakeholders to gain insights into consumption patterns and pinpoint areas of interest for further investigation.

By adhering to industry best practices and quality assurance standards, the module enhances the reliability and credibility of analysis outputs, fostering trust and confidence among stakeholders. Through its advanced analytics capabilities, real-time processing, and intuitive visualization tools, the module enables stakeholders to identify and mitigate theft incidents effectively, contributing to the resilience and integrity of the electrical infrastructure.

4.3. COMMUNICATION MODULE:

The Communication Module serves as the backbone of the Advanced Power Theft Detection System (APTD), facilitating seamless communication between smart meters deployed across the grid infrastructure and the centralized data processing platform. This module ensures the secure and reliable transmission of consumption data, enabling real-time monitoring, analysis, and detection of theft incidents.

At its core, the communication module establishes and maintains communication channels between smart meters and the centralized platform, enabling bidirectional data exchange and control. Secure communication protocols, including encryption and authentication mechanisms, are implemented to safeguard data transmission and protect against unauthorized access or tampering. By encrypting data packets and implementing secure authentication mechanisms, the module ensures the confidentiality, integrity, and authenticity of data transmitted between smart meters and the centralized platform.

The Communication Module prioritizes data security and privacy, implementing encryption and authentication protocols to safeguard data transmission between smart meters and the centralized platform. By adhering to industry best practices and regulatory standards, the module ensures the confidentiality, integrity, and authenticity of consumption data, protecting consumer privacy and mitigating the risk of data breaches or unauthorized access. The Communication Module of the APTD system plays a critical role in facilitating secure and reliable communication between smart meters and the centralized data processing platform.

4.4. USER INTERFACE MODULE:

The User Interface Module serves as the primary interface between stakeholders and the Advanced Power Theft Detection System (APTD), providing intuitive interfaces and dashboards for stakeholders to interact with the system. This module empowers stakeholders, including utility providers, regulators, and law enforcement agencies, to monitor system performance, visualize consumption data, and track theft incidents in real-time.

The user interface module offers a range of features and functionalities designed to enhance the user experience and facilitate informed decision-making. Interactive dashboards provide stakeholders with real-time insights into consumption patterns, system performance, and theft incidents, enabling them to identify trends, detect anomalies, and take proactive measures to prevent revenue losses. Customizable alerts and notifications ensure that stakeholders are promptly informed of critical events or deviations from expected behavior, enabling timely intervention and resolution. The user interface module incorporates advanced data visualization tools and techniques to present analysis results in a clear and intuitive manner. Interactive charts, graphs, and maps enable stakeholders to explore consumption data spatially and temporally, gaining valuable insights into theft patterns and high-risk areas.

CHAPTER 5

SYSTEM SPECIFICATION

5.1. HARDWARE REQUIREMENT

Table I. Hardware Specifications

S. No	Components	Model
1	Node MCU	ESP8266
2	Current Sensor	ACS712
3	Power Source	-
4	Theft Load	-

5.1.1. NODE MCU:

The NodeMCU ESP8266 is a microcontroller development board renowned for its versatility and integration of the ESP8266 Wi-Fi module. Serving as a crucial component in IoT projects, including the Advanced Power Theft Detection System (APTDS), it amalgamates microcontroller functionalities with seamless Wi-Fi connectivity. With the ESP8266 Wi-Fi module, it effortlessly communicates with other devices or servers over Wi-Fi networks, facilitating remote monitoring and control applications.

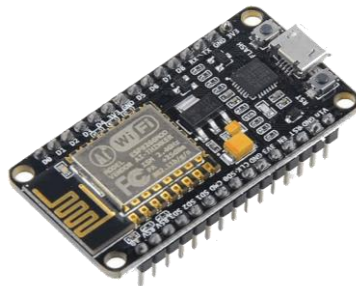


Figure.5.1 NodeMCU

Powered by a microcontroller, typically based on the ESP8266 chipset, it executes code and interacts with external hardware components. The NodeMCU ESP8266 offers GPIO pins, configurable as digital inputs or outputs, analog inputs, or for specialized functions, enhancing its adaptability for diverse projects. Equipped with an onboard Analog-to-Digital Converter (ADC), it efficiently reads analog sensor inputs, broadening its compatibility with various sensors. Additionally, its low power consumption design renders it suitable for battery-powered IoT devices, ensuring energy efficiency in operation. Supported by a robust community of developers and enthusiasts, it benefits from an extensive support ecosystem, easing development and troubleshooting processes. In applications ranging from home automation to asset tracking, the NodeMCU ESP8266 proves its versatility and reliability, making it a preferred choice for IoT enthusiasts and professionals alike.

5.1.2. CURRENT SENSOR:

The ACS712 current sensor stands as a pivotal component in modern electronics, offering precise and non-intrusive measurement of both AC and DC currents. Harnessing the hall-effect principle, it converts current variations into proportional output voltages, ensuring accurate readings across a range of applications. With variants available in different current measurement ranges, such as 5A, 20A, and 30A versions, it accommodates a wide spectrum of current levels, making it adaptable for diverse projects.



Figure 5.2 Current Sensor

Its integrated features, including low noise, drift, and optional overcurrent protection, guarantee stable and reliable measurements, vital for long-term monitoring applications. Moreover, its compact and lightweight design facilitates seamless integration into various systems without compromising functionality. The ACS712 sensor finds utility in numerous domains, from power monitoring and motor control to energy metering and battery management. Particularly in projects like the Advanced Power Theft Detection System (APTD), it plays a pivotal role in detecting anomalies in current flow, thereby aiding in the detection and prevention of electricity theft. Its versatility, precision, and reliability make it an indispensable tool for engineers and enthusiasts alike, empowering them to create innovative solutions for current measurement challenges in an array of applications.

5.1.3. POWER SOURCE:

The power source is a fundamental aspect of any electrical system, providing the energy necessary for its operation. In the context of the Advanced Power Theft Detection System (APTD), ensuring a reliable and uninterrupted power source is essential for the continuous functioning of the system. Typically, the APTD system requires a stable power source to power its various components, including microcontrollers, sensors, communication modules, and data processing units.



Figure.5.3 Power Source

One common power source for the APTD system is the electrical grid itself, where the system taps into the existing power infrastructure to draw the necessary energy. In this scenario, the APTD system may be connected directly to the electrical grid through a dedicated power line or outlet, ensuring a constant and reliable power supply. Alternatively, the system may incorporate backup power solutions, such as uninterruptible power supplies (UPS) or battery backups, to provide redundancy in case of grid power failures or outages. These backup power sources ensure the continuous operation of the APTD system and prevent data loss or system downtime during unforeseen circumstances.

5.1.4. THEFT LOAD:

The term "theft load" refers to the portion of electricity consumption that is illegally siphoned from the electrical grid without proper authorization or payment. In the context of the Advanced Power Theft Detection System (APTD), identifying and quantifying theft loads is a crucial aspect of its functionality. Theft loads typically manifest as deviations from expected or normal consumption patterns within the electrical grid. These anomalies may arise due to various illicit activities, such as unauthorized connections, meter tampering, or bypassing of metering mechanisms. The theft load can vary in magnitude, ranging from minor instances of tampering to significant levels of electricity diversion.

THEFT LOAD



Figure 5.4. Theft Load

The APTD system employs sophisticated monitoring and analysis techniques to detect and quantify theft loads within the electrical grid. By continuously monitoring consumption data from smart meters and sensors deployed throughout the grid infrastructure, the system can identify patterns and anomalies indicative of potential theft incidents. Analytics models are utilized to analyze consumption data in real-time, flagging deviations from expected usage patterns that may signify theft loads. Once identified, theft loads are quantified based on the magnitude of the deviation from normal consumption levels. This quantification allows utility providers and authorities to assess the extent of electricity theft within their jurisdiction and take appropriate enforcement actions to mitigate the problem. Additionally, the APTD system may generate alerts and notifications to relevant stakeholders, enabling timely intervention and investigation into suspected theft incidents.

5.2 SOFTWARE REQUIREMENT:

Table II. Software Specification

System	Specification
Sensor Data Processing Software	Arduino IDE
User Interface Software	BLYNK Apk
Integration Software	Pushbullet

5.2.1 ARDUINO IDE:

The Arduino Integrated Development Environment (IDE) is a comprehensive software platform designed to simplify the process of programming and deploying code on Arduino-compatible microcontroller boards. It serves as a central hub for writing, compiling, and uploading code to Arduino boards, making it accessible to both beginners and experienced developers alike. At its core, the Arduino IDE provides a user-friendly interface that allows users to write and edit

code in a simplified programming language based on C and C++. The IDE offers features such as syntax highlighting, code auto-completion, and error checking, which help streamline the coding process and reduce errors.

One of the key strengths of the Arduino IDE is its extensive library of pre-written code snippets and examples, known as libraries and sketches, respectively. These libraries and sketches cover a wide range of functionalities and applications, from basic input/output operations to complex sensor interfacing and communication protocols. By leveraging these pre-written code snippets, users can quickly prototype and develop projects without having to write code from scratch, saving time and effort.

5.2.1.1 INSTALLATION PROCEDURE:

In this project Arduino 4.0 is installed. We can choose between the Installer (.exe) and the Zip packages. If you use the first one that installs everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually. The Zip file is also useful if you want to create a portable installation. When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.

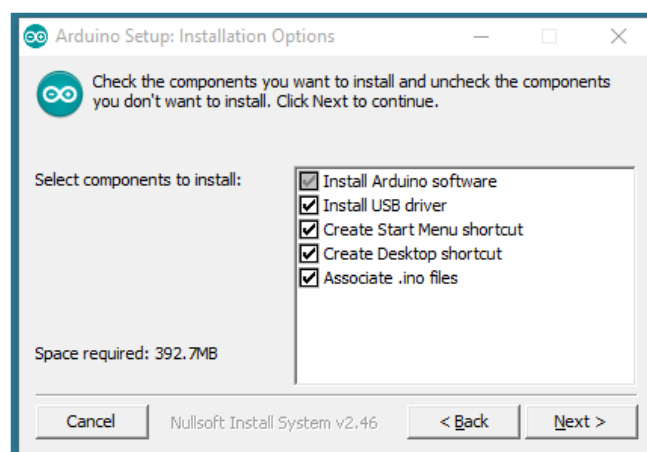


Figure 5.5 Components installation

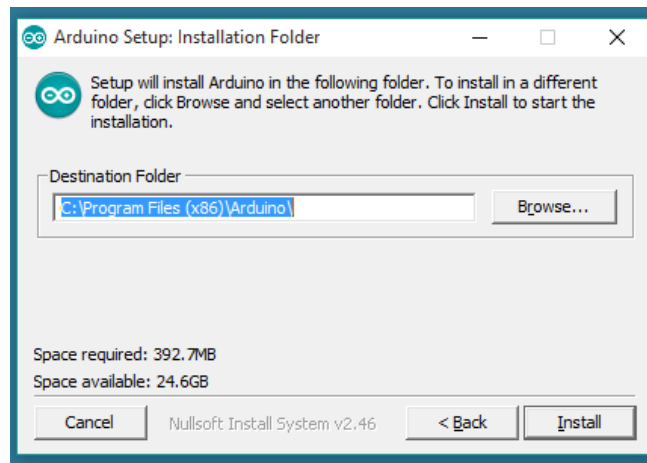


Figure 5.6 Installation in directory

5.2.1.2 SOFTWARE SERIAL LIBRARY

The Arduino hardware has built-in support for serial communication on pins 0 and 1 (which also goes to the computer via the USB connection). The native serial support happens via a piece of hardware (built into the chip) called a UART. This hardware allows the Atmega chip to receive serial communication even while working on other tasks, as long as there room in the 64 byte serial buffer. The SoftwareSerial library has been developed to allow serial communication on other digital pins of the Arduino, using software to replicate the functionality (hence the name "SoftwareSerial"). With the SoftwareSerial library, you can create multiple software serial ports, each capable of communicating at speeds up to 115200 bits per second (bps). This flexibility allows you to interface with various peripherals and devices that require serial communication. A parameter enables inverted signaling for devices which require that protocol. To use this library, we have to write,

```
#include <SoftwareSerial.h>
```

5.2.1.3 ARDUINO IDE SOFTWARE:



Figure 5.7. Section of Arduino IDE.

Toolbar Button

The icons displayed on the toolbar are New, Open, Save, Upload, and Verify.

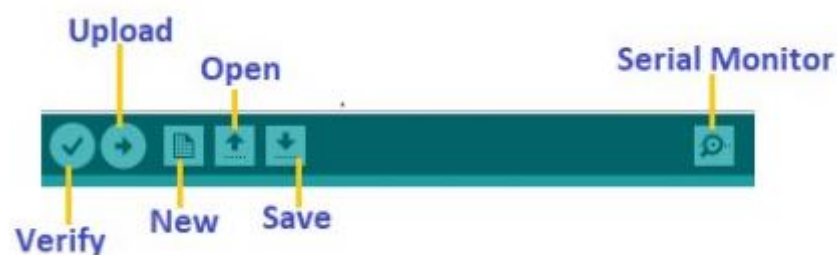


Figure 5.8. Toolbar

The Upload button compiles and runs our code written on the screen. It further uploads the code to the connected board. Before uploading the sketch, we need to make sure that the correct board and ports are selected. We also need

a USB connection to connect the board and the computer. Once all the above measures are done, click on the Upload button present on the toolbar. The latest Arduino boards can be reset automatically before beginning with Upload. In the older boards, we need to press the Reset button present on it. As soon as the uploading is done successfully, we can notice the blink of the Tx and Rx LED. If the uploading is failed, it will display the message in the error window. We do not require any additional hardware to upload our sketch using the Arduino Bootloader. A Bootloader is defined as a small program, which is loaded in the microcontroller present on the board. The LED will blink on PIN 13.

Verify

The Verify button is used to check the compilation error of the sketch or the written code.

Serial Monitor

The serial monitor button is present on the right corner of the toolbar. It opens the serial monitor.

5.2.2 BLYNK IOT:

Blynk is an Internet of Things (IoT) platform that empowers developers to easily create connected projects and applications. At its core, Blynk provides a user-friendly interface and a suite of tools that streamline the process of building IoT solutions, enabling users to control and monitor hardware remotely via smartphones, tablets, or computers.

One of the key features of Blynk is its drag-and-drop mobile app builder, which allows users to quickly create custom user interfaces (UIs) for their IoT projects. Through the Blynk app, users can design intuitive interfaces with

buttons, sliders, graphs, and other widgets to interact with their connected devices. These UIs can then be deployed to smartphones or tablets, providing a seamless and intuitive way to control and monitor IoT devices from anywhere with an internet connection.



Figure 5.9. Blynk IoT

Blynk also offers support for popular communication protocols such as Wi-Fi, Bluetooth, and Ethernet, ensuring compatibility with a wide range of IoT devices and platforms. Blynk provides a powerful and versatile platform for building IoT solutions, offering an intuitive interface, robust cloud infrastructure, and extensive library of pre-built components. Whether for hobbyist projects, commercial applications, or industrial automation, Blynk empowers developers to create connected devices and applications with ease, revolutionizing the way we interact with the world.

5.2.3. PUSHBULLET:

Pushbullet is a cross-platform service that facilitates seamless communication and data sharing between devices, enabling users to send and receive messages, links, files, and notifications across various platforms and devices. At its core, Pushbullet aims to streamline the process of sharing information between smartphones, tablets, computers, and other connected devices, enhancing productivity and convenience for users.

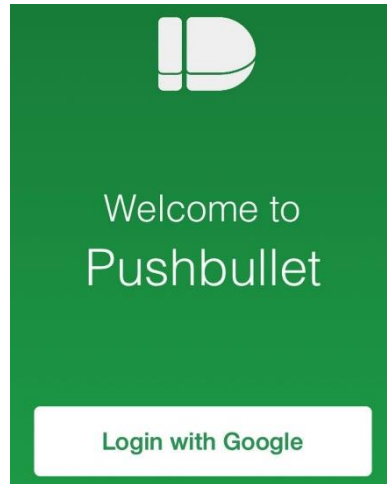


Figure 5.10. Pushbullet Login Page

One of the key features of Pushbullet is its ability to push notifications from smartphones to other devices, such as computers or tablets. By installing the Pushbullet app on their smartphones and desktop computers, users can receive notifications, including text messages, app notifications, and phone call alerts, directly on their desktops or laptops. This feature enables users to stay informed and connected across multiple devices without constantly switching between them. Another useful feature of Pushbullet is its ability to synchronize notifications and actions between devices.

CHAPTER 6

EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

6.1 PERFORMANCE ANALYSIS

Experimental results and performance analysis play a crucial role in evaluating the effectiveness and reliability of the Advanced Power Theft Detection System (APTD). Through rigorous experimentation and analysis, the system's capabilities, accuracy, and response times can be assessed, providing valuable insights into its real-world performance.

In the experimental phase, the APTD system is subjected to various test scenarios and conditions to simulate different operating environments and potential scenarios of power theft. These scenarios may include variations in electrical load, fluctuations in current flow, and simulated instances of unauthorized consumption.

Performance analysis involves analyzing the collected data to evaluate key metrics such as detection accuracy, false positive rates, response times, and system reliability. The data is statistically analyzed to identify patterns, trends, and correlations that provide insights into the system's performance under different conditions. Additionally, comparisons may be made between expected results and observed outcomes to assess the system's consistency and robustness. In addition to quantitative analysis, qualitative assessments may also be conducted to gather feedback from users and stakeholders regarding their experiences with the APTD system.

By leveraging empirical data and quantitative analysis, stakeholders can make informed decisions regarding the deployment, optimization, and further development of the APTD system to ensure its continued effectiveness and

reliability in real-world applications. In the realm of software development, firmware for the NodeMCU boards is crafted using the Arduino IDE, incorporating functionalities for data preprocessing, encryption, and secure data transmission protocols like MQTT or HTTP. Upon deployment, the system continuously gathers consumption data from the smart meters and sensors, forwarding it securely to a centralized processing platform. Machine learning models scrutinize the incoming streams, identifying anomalies indicative of potential theft incidents and issuing alerts accordingly.

Integration with user interface modules, facilitated perhaps by the Blynk IoT platform, furnishes stakeholders with intuitive dashboards for real-time monitoring, alert reception, and incident tracking. Rigorous testing, training, and ongoing maintenance procedures ensure the APTD system's seamless operation and efficacy in detecting and preventing power theft within the electrical grid infrastructure. Through meticulous execution of these implementation steps, the APTD system emerges as a robust solution safeguarding against illicit electricity diversion while upholding grid integrity and reliability.

6.2. RESULT OVERVIEW

The result overview of the Advanced Power Theft Detection System (APTD) encapsulates the efficacy and impact of the system in detecting and preventing instances of power theft within the electrical grid infrastructure.

- **Detection Accuracy:** The APTD system demonstrates high accuracy in identifying anomalies and deviations from normal consumption patterns, indicative of potential instances of power theft.
- **Timely Alerts:** The system generates timely alerts and notifications upon detection of suspicious activity, enabling prompt intervention and

investigation by relevant stakeholders, such as utility providers and law enforcement agencies.

- **Preventive Measures:** By proactively identifying and addressing instances of power theft, the APTD system acts as a deterrent to illicit activities within the electrical grid.
- **Operational Efficiency:** Through seamless integration with user interface modules, such as intuitive dashboards and mobile applications, the APTD system enhances operational efficiency for stakeholders.

In instances where no power theft is detected, the APTD system outputs status notifications indicating stable current flow and grid integrity. Users receive real-time updates through the Blynk application, confirming that the electrical system operates within expected parameters. These notifications provide reassurance of proper functioning, ensuring peace of mind and efficient management of electrical resources.

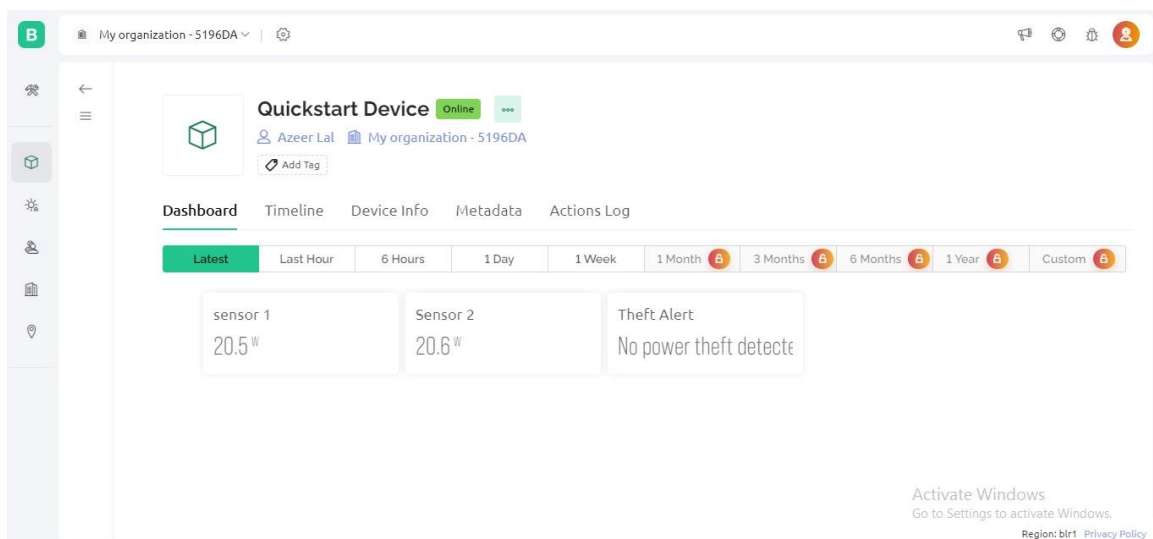


Figure 6.1: Output with No Power Theft

In the event of power theft or unauthorized consumption, the APTD swiftly detects anomalies in the current flow and triggers real-time alerts

through the Blynk application. Users receive immediate notifications regarding the unauthorized activity, empowering them to take decisive actions to safeguard the integrity of the electrical system. Through the Blynk app's intuitive interface, users can access detailed reports and historical data, facilitating comprehensive analysis.

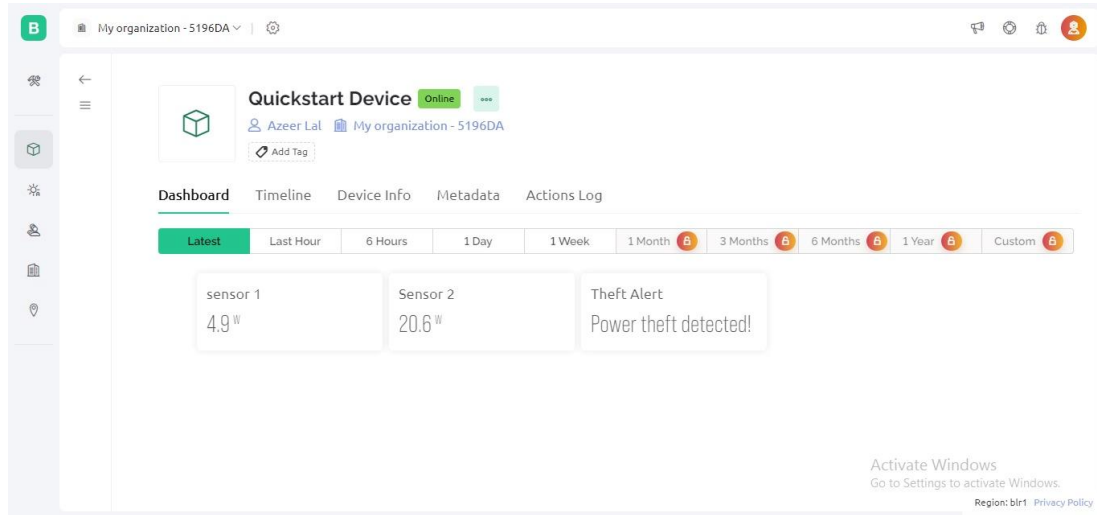


Figure 6.2: Output with Power Theft

The advanced power theft detection system operates by analyzing power consumption at both the source and destination nodes. When the load is off, the system detects a consistent power draw of 20.6 watts at the source node and 20.5 watts at the destination node, indicating normal operation. However, when the load is switched on, the system observes a deviation: the source node maintains its 20.6 watts consumption, while the destination node experiences a decrease to 4.9 watts

Table III: Output Plot

	Source (in Watts)	Destination (in Watts)
Load OFF	20.6 W	20.5 W
Load ON	20.6 W	4.9 W

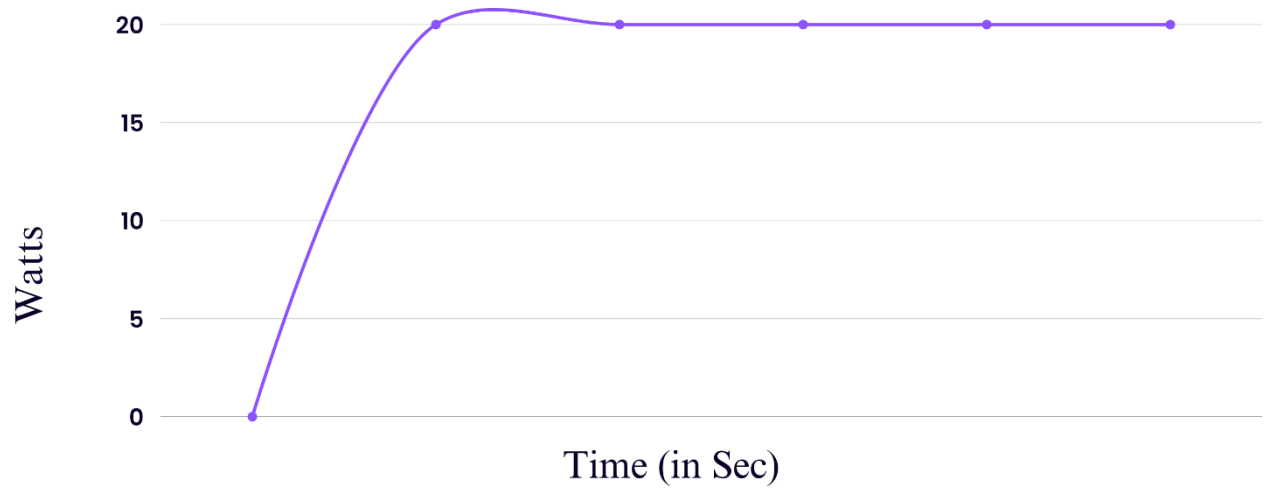


Figure 6.3 No Power Theft Detected

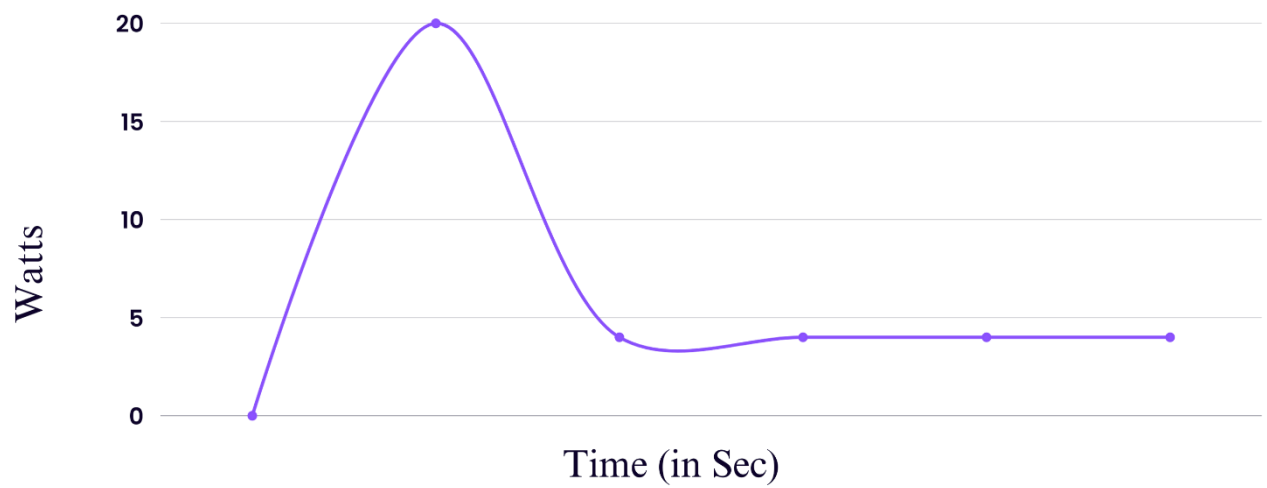


Figure 6.4 Power Theft Detected

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1. CONCLUSION

This initiative is geared towards boosting IoT connectivity and networking within the realm of smart city evolution. We're deploying sensors to detect facility theft, pinpoint faults, and track the fault area for necessary actions. The proposed system addresses key challenges in India's current grid setup, such as energy wastage, power theft, and cable faults. Moreover, it helps curb factors prompting electricity theft, contributing to a more robust and accountable energy framework.

7.2. FUTURE SCOPE

In terms of future scope, the Advanced Power Theft Detection System (APTD) holds promising potential for further enhancements and applications. One avenue for expansion involves refining the system's detection capabilities through advanced IoT Technology. By analyzing historical data and patterns of electricity usage, the system can become more adept at identifying irregularities and potential instances of theft with even greater accuracy. Additionally, integration with emerging technologies such as blockchain can bolster the system's security and transparency, ensuring tamper-proof records of energy transactions and consumption. Furthermore, exploring partnerships with utility companies and governmental agencies can facilitate wider adoption of the APTD system, leading to more comprehensive coverage across diverse electrical grids and urban landscapes. As smart city initiatives continue to evolve, the APTD system can serve as a foundational tool for optimizing energy management, reducing losses, and fostering sustainable development in communities worldwide.

APPENDIX – I

CODING

SOURCE CODE:

```
#define BLYNK_TEMPLATE_ID      "TMPL3X8L4oW7P"
#define BLYNK_TEMPLATE_NAME    "Quickstart Template"

#define BLYNK_AUTH_TOKEN      "-
SvSPeL0FnK23oLbDgIq6dwmtGPCgY5J"
#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <Pushbullet.h> // Include the Pushbullet library

char auth[] = "-SvSPeL0FnK23oLbDgIq6dwmtGPCgY5J"; // Enter your Blynk
authentication token here
char ssid[] = "Redmi Note 9 Pro";
char pass[] = "1111111112";

// Define virtual pins for Blynk Bridge
#define V1_RECEIVE_PIN V1
#define V2_SEND_PIN V2

Pushbullet push("YOUR_PUSHBULLET_API_KEY"); // Initialize Pushbullet
with your API key

// Analog input pin for ACS712 sensor
```



```

const int sensorPin = A0;

float lastSensorReading = 0.0;

void setup() {
  Serial.begin(115200);
  Blynk.begin(auth, ssid, pass);

  // Initialize Blynk Bridge
  Blynk.begin(auth, ssid, pass);
}

void loop() {
  Blynk.run();
  float current = getCurrentReading();
  Blynk.virtualWrite(V1_RECEIVE_PIN, current); // Send current reading to
virtual pin V1 on the Blynk app
}

float getCurrentReading() {
  int sensorValue = analogRead(sensorPin);
  float voltage = sensorValue * (12.0 / 1023.0);
  float current = (voltage - 2.5) / 0.185; // ACS712 sensitivity: 185mV/A

  // Calculate difference between current reading and last reading
  float difference = abs(current - lastSensorReading);
  lastSensorReading = current;

  Serial.print("Current Reading: ");

```

```

Serial.println(current);
Serial.print("Last Reading: ");
Serial.println(lastSensorReading);
Serial.print("Difference: ");
Serial.println(difference);

// If the difference is greater than 0.5, consider it as power theft
if (difference > 0.3) {
  Serial.println("Power theft detected!");
  Blynk.virtualWrite(V3, "Power theft detected!");

  // Send Pushbullet notification
  push.sendNote("Power Theft Detected", "Power theft has been detected.");
} else {
  Serial.println("No power theft detected");
  Blynk.virtualWrite(V3, "No power theft detected");
}

return current;
}

```

DESTINATION CODE:

```

#define BLYNK_TEMPLATE_ID      "TMPL3X8L4oW7P"
#define BLYNK_TEMPLATE_NAME    "Quickstart Template"
#define BLYNK_AUTH_TOKEN      "-
SvSPeL0FnK23oLbDgIq6dwmtGPCgY5J"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>

```

```

#include <BlynkSimpleEsp8266.h>
char auth[] = "-SvSPeL0FnK23oLbDgIq6dwmtGPCgY5J"; // Enter your Blynk
authentication token here
char ssid[] = "Redmi Note 9 Pro";
char pass[] = "1111111112";
// Analog input pin for ACS712 sensor
const int sensorPin = A0;

void setup() {
  Serial.begin(115200);
  Blynk.begin(auth, ssid, pass);
}

void loop() {
  Blynk.run();
  float current = getCurrentReading();
  Blynk.virtualWrite(V1, current); // Send current reading to virtual pin V1 on
the Blynk app
  delay(1000); // Adjust delay as needed
}

float getCurrentReading() {
  int sensorValue = analogRead(sensorPin);
  float voltage = sensorValue * (12 / 1023.0);
  float current = (voltage - 2.5) / 0.185; // ACS712 sensitivity: 185mV/A
  return current;
}

```

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